Chapter 2 — Discretisation

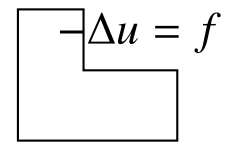
Galerkin and spectral Galerkin methods

MTH3340 · Numerical Methods for PDEs

1 Introduction

The variational problem (weak form) gives us

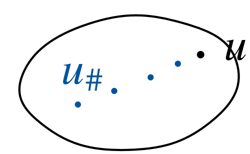
- no strong (pointwise) sense needed,
- $\exists!$ of solutions,
- but on an ∞-dimensional space, e.g., $H^1_0(\Omega)$.



$$u \in V \;:\; \int_\Omega \nabla u \cdot \nabla v = \int_\Omega f v \;\; \forall v \in V_0 \qquad \xrightarrow{\text{FEM}} \qquad \boldsymbol{A}\vec{\boldsymbol{u}}_\# = \vec{\boldsymbol{b}},$$

a finite-dimensional linear system: only FLOPs $(+, -, \times, \div)$ in $\mathbb{R}$ — a computer can do it!

The price: $u_\# \approx u$ is an approximation. The deal we aim for: $u_\# \to u$ as the cost grows (FEM, spectral Galerkin, . . .).



2 Galerkin method

The variational problem (Chapter 1),

$$u \in V_g \ : \ a(u,v) = \ell(v) \quad \forall v \in V_0, \qquad V_g \doteq \{u \in V \ : \ u = g \text{ on } \Gamma_0\}$$

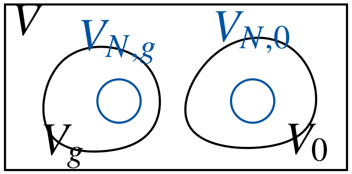
(for us, $V = H^1(\Omega)$, $V_g = H^1_g(\Omega)$, $V_0 = H^1_0(\Omega)$).

Definition (Galerkin approximation)

Consider finite-dimensional subspaces $V_{N,0} \subset V_0$ and $V_N \subset V$, with $N \doteq \dim(V_N) < \infty$, and the affine space $V_{N,g} \subset V_N$ of discrete functions satisfying the Dirichlet bc's. The Galerkin approximation reads

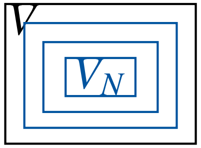
$$u_N \in V_{N,g} \ : \ a(u_N, v_N) = \ell(v_N) \quad \forall v_N \in V_{N,0}.$$

Well-posed by the finite-dimension theorem of Chapter 1, and $\Longleftrightarrow$ a linear system $A\vec{\mu} = \vec{f}$ (below).



Bases and coefficient vectors

"$\lim_{N\to\infty} V_N = V$" (in a sense to be made precise), and $N \leftrightarrow$ computational cost.



Definition (Basis of a finite dimensional vector space)

Given a finite dimensional real vector space V_M, the set $\{b_1, \dots, b_M\} \subset V_M$, $M \in \mathbb{N}$, is a basis of V_M if for all $v \in V_M$ there is a *unique* set of coefficients $\{v_i\}_{i=1}^{M} \subset \mathbb{R}$ such that $v = v_1 b_1 + \cdots + v_M b_M$. M is the dimension of V_M.

An *ordered* basis gives an isomorphism $V_M \cong \mathbb{R}^M$[1] (as in the finite-dimension proofs of Chapter 1!),

$$V_M \ni v(x) = v_1 b_1(x) + \cdots + v_M b_M(x) \quad \longleftrightarrow \quad \vec{v} = [v_1, \dots, v_M] \in \mathbb{R}^M.$$

[1] An *isomorphism* of vector spaces is a linear bijective map $\Phi : V_M \to \mathbb{R}^M$. *Linear* means $\Phi(\alpha u + \beta v) = \alpha\Phi(u) + \beta\Phi(v)$. *Bijective* means *one-to-one* (distinct elements have distinct images) and *onto* (every vector in $\mathbb{R}^M$ is the image of some $v \in V_M$); such a map can be inverted.

From Galerkin to a linear system

Definition (Galerkin method with offset function)

Pick a discrete offset $u_{N,0} \in V_{N,g}$. The Galerkin solution reads $u_N = u_{N,0} + \delta u_N$, where

$$\delta u_N \in V_{N,0} \;:\; a(\delta u_N, v_N) = \tilde{\ell}(v_N) \doteq \ell(v_N) - a(u_{N,0}, v_N) \quad \forall v_N \in V_{N,0}.$$

Lemma (Galerkin system as linear system)

Consider a basis $\mathcal{B} \doteq \{b_N^1, \ldots, b_N^{N_0}\}$ *of* $V_{N,0}$*, where* $N_0 \doteq \dim(V_{N,0})$. *The Galerkin problem above is equivalent to the linear system: find* $\delta u_N = \sum_{i=1}^{N_0} \mu_i b_N^i$ *with*

$$\vec{\boldsymbol{\mu}} \in \mathbb{R}^{N_0} \;:\; \boldsymbol{A}\vec{\boldsymbol{\mu}} = \vec{\boldsymbol{f}}, \qquad \boldsymbol{A}_{ij} \doteq a(b_N^j, b_N^i), \quad \vec{\boldsymbol{f}}_i \doteq \tilde{\ell}(b_N^i).$$

Same offset trick as in Chapter 1, now at the discrete level: all the work happens in the vector space $V_{N,0}$.

Proof: testing against a basis is enough

Proof. Write $\delta u_N = \sum_j \mu_j b_N^j$ (unique coefficients). Testing the Galerkin equation with each basis function $v_N = b_N^i$,

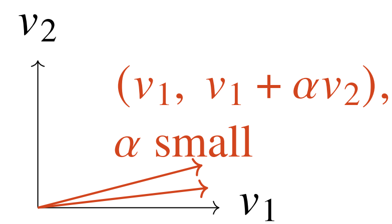
$$a\Big(\sum_{j=1}^{N_0} \mu_j b_N^j,\ b_N^i\Big) = \sum_{j=1}^{N_0} \underbrace{a(b_N^j, b_N^i)}_{\boldsymbol{A}_{ij}} \mu_j = \tilde{\ell}(b_N^i) = \vec{\boldsymbol{f}}_i, \qquad i = 1, \dots, N_0,$$

so the coefficients of δu_N solve the linear system. Conversely, any test function is $v_N = \sum_i v_i b_N^i$; multiplying equation i by v_i and adding up, the variational equation holds for v_N. $\square$

By linearity, testing against a basis $\iff$ testing against all of $V_{N,0}$. We will use this argument all the time.

Does the basis matter?

The choice of $\mathcal{B}$ does not affect the solution u_N, but it does affect $\boldsymbol{A}$, $\vec{\boldsymbol{f}}$ and $\vec{\boldsymbol{\mu}}$; it fixes the isomorphism $V_{N,0} \leftrightarrow \mathbb{R}^{N_0}$.



In floating point arithmetic it matters a lot: the matrix can be well or ill conditioned.

Compare the bases

$$(v_1, v_2) \qquad \text{vs.} \qquad (v_1,\ v_1 + \alpha v_2),\ \alpha > 0 \text{ small:}$$

the second is nearly linearly dependent $\longrightarrow$ ill-conditioned $\boldsymbol{A}$, rounding errors blow up.

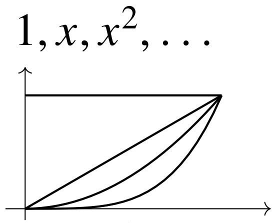
3 Spectral Galerkin method

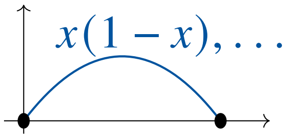
$\Omega = [a, b]$. Approximate V (and V_0) with the polynomials up to a given order (vanishing on the boundary, for V_0):

$$V_N = \operatorname{span}\{1, x, x^2, \ldots, x^p\} = \mathcal{P}_p(\mathbb{R}), \qquad \dim(V_N) = p + 1,$$

$$V_{N,0} = \mathcal{P}_p(\mathbb{R}) \cap C_0^0(\bar{\Omega}), \qquad \dim(V_{N,0}) = p - 1 \quad (p \geq 2);$$

e.g., on $[0, 1]$, $V_{N,0} = \operatorname{span}\{x(1-x),\ x^2(1-x), \ldots,\ x^{p-1}(1-x)\}$.





Monomial-type bases are ill-conditioned (Vandermonde-like matrices); in practice one uses, e.g., Legendre polynomials (MTH2051/3051).

Computing the entries

With the (bc-free, for a moment) basis $\mathcal{B} = \{1, x, x^2, \ldots\}$, the system entries are integrals,

$$\boldsymbol{A}_{ij} = \int_\Omega \partial_x b^i \, \partial_x b^j, \qquad \vec{\boldsymbol{f}}_j = \int_\Omega f \, b^j.$$

E.g.,

$$\boldsymbol{A}_{22} = \int_\Omega \partial_x x^2 \cdot \partial_x x^2 = \int_\Omega 2x \cdot 2x = 4 \int_\Omega x^2.$$

And the solution vector gives back a function,

$$\vec{\boldsymbol{\mu}} = [2, 4, 3] \quad \longleftrightarrow \quad u_N(x) = 2 \cdot 1 + 4\,x + 3\,x^2.$$

Q: How do we compute $\int_\Omega f \, b^j$ for a general f, with FLOPs only?

Numerical integration.

Numerical integration: quadratures

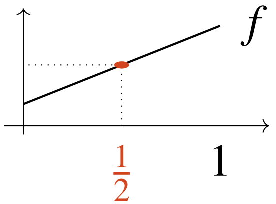
Definition (m-point quadrature)

On $[a,b] \subset \mathbb{R}$ and for $m \in \mathbb{N}$, an m-point quadrature is an approximation

$$\int_a^b f(x)\,\mathrm{d}x \approx Q_m^{[a,b]}(f) \doteq \sum_{j=1}^{m} \omega_j^m\, f(\xi_j^m),$$

defined by the *weights* ω_j^m and the *points* ξ_j^m.

The m-point Gauss quadrature is exact for polynomials up to degree $2m-1$. E.g., midpoint ($m=1$): $f(\frac{1}{2}) = \int_0^1 f$ for linear f.



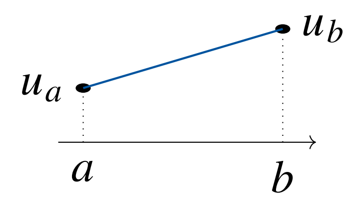
Points and weights are tabulated once on the reference interval $[-1,1]$; a change of variables does the rest:

$$\int_a^b f(x)\,\mathrm{d}x = \frac{b-a}{2}\int_{-1}^{1} f\Big(\frac{a+b}{2} + \frac{b-a}{2}\,\xi\Big)\mathrm{d}\xi.$$

Exactness degree $\Longrightarrow$ accuracy: see the lecture notes.

Non-homogeneous bc's

$u(a) = u_a$, $u(b) = u_b$: we need a discrete offset in V_N through both points — the linear interpolant ($p \geq 1$ suffices),



$$u_{p,0}(x) = \frac{bu_a - au_b}{b-a} + \frac{u_b - u_a}{b-a} x.$$

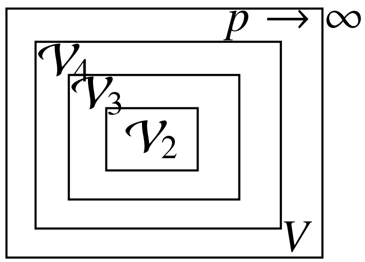
Definition (Spectral Galerkin method with offset function)

For $\Omega = [a,b]$, the bc's above and the order $p \in \mathbb{N}^+$, let $\mathcal{V}_p \doteq \mathcal{P}_p([a,b]) \cap C^0_0([a,b])$ and compute

$$\delta u_p \in \mathcal{V}_p \; : \; a(\delta u_p, v_p) = \ell(v_p) - a(u_{p,0}, v_p) \quad \forall v_p \in \mathcal{V}_p.$$

The spectral Galerkin approximation reads $u_p \doteq u_{p,0} + \delta u_p$.

A family of spaces



We have defined not one space but a family $\{\mathcal{V}_p\}_{p=1}^{\infty}$, and with it a family of solutions,

$$u_2,\ u_3,\ u_4,\ \ldots\ \longrightarrow\ u\ ?$$

$$\text{“}\lim_{p\to\infty} \|u - u_p\|_V = 0\text{”}$$

Does it converge? How fast (vs. computational cost)? That is the error analysis of Galerkin methods — coming next.